

When Bat Speed Isn't the Whole Story

Cam Smith, the power/contact trade-off, and why raw correlations can hide the metric that actually matters.

#2

bat speed gainer in
MLB, 2025→2026

P8

squared-up%
percentile — bottom
decile

−.012

wOBA change
despite the bat speed
gain

Elite Bat Speed, Declining Production

Cam Smith (Houston Astros, RF, age 23) posted one of the largest bat speed gains in Major League Baseball from 2025 to 2026 — a jump from the 84th to the **98th percentile**. By the surface-level logic of "hit it harder, do more damage," 2026 should have been a breakout year. Instead, his actual production *declined*.

Metric (MLB percentile)	2025	2026	Δ
Bat Speed	84	98	+14
Squared-Up%	3	10	+7 (still bottom decile)
Barrel%	6.9%	11.8%	+71% relative
xwOBA	.304	.334	+.030
wOBA (actual)	.298	.286	−.012
Ground Ball%	45.1%	47.2%	+2.1

THE CORE FINDING

Bat speed jumped 14 percentile points. Squared-up% — a measure of how efficiently that speed is converted into quality contact — barely moved, staying in the bottom decile of MLB both years. Underlying contact quality improved (xwOBA +.030), but actual outcomes got worse (wOBA −.012), because harder swings without proportional improvements in barrel timing produce more mishits, not more damage.

Why this matters beyond one player

This case is a live test of a principle borrowed from force plate methodology (VALD's *Practitioner's Advanced Guide to Force Plates*): a single "peak" metric — jump height, peak force, or here, bat speed — can improve while the underlying movement quality that determines outcome stays flat or even regresses. The guide's own words apply almost verbatim to hitting: "if everything is better to the same degree, you have a more athletic group; if one metric separates while others stay flat, you're seeing a quality that truly differentiates." Bat speed separated. Squared-up% didn't. That divergence is the signal.

The Power/Contact Trade-off

Across 205 qualified MLB hitters (2026 season, Statcast bat-tracking leaderboard), bat speed and squared-up% are **not independent** — they are negatively correlated with each other ($r = -0.65$). Hitters who swing harder tend, on average, to square the ball up less often. This is a real, structural feature of hitting mechanics, not statistical noise: an independent sample of 177 additional qualified hitters produced an almost identical correlation ($r = -0.665$).

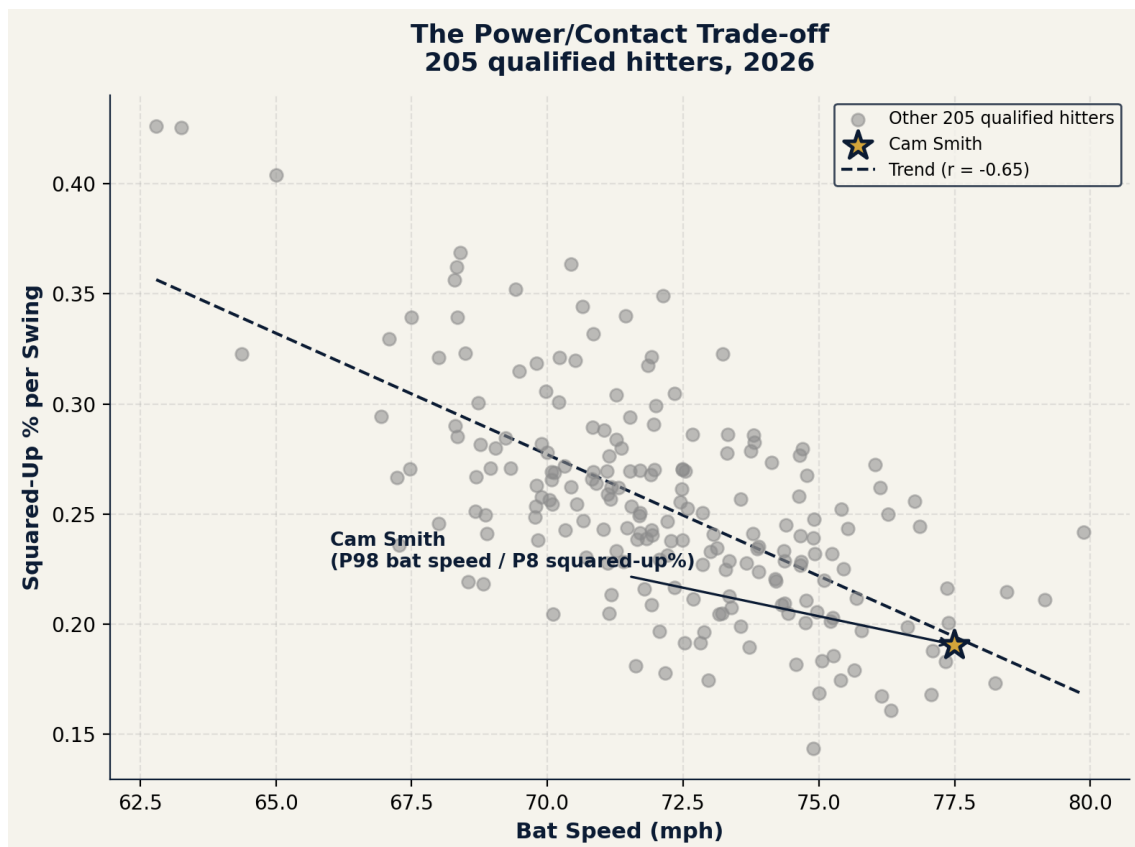


Figure 1. Bat speed vs. squared-up% across 205 qualified hitters (2026). Cam Smith sits at the extreme end of the trade-off — elite bat speed, bottom-decile squared-up%.

Cam Smith isn't an outlier from the pattern — he's a clean illustration of it, sitting almost exactly on the trend line at the far end of the bat speed axis. His profile isn't unusual mechanically; it's the expected trade-off taken to its current extreme.

03 — THE STATISTICAL CORRECTION

Why the Simple Correlation Is Misleading

Because bat speed and squared-up% move in opposite directions across the population, looking at squared-up%'s *simple* correlation with outcome badly understates its importance — a classic statistical suppression effect. In the 205-player dataset:

	Raw correlation with xwOBA	Standardized coefficient (controlling for other 3 features)
Bat Speed	$r = 0.449$	+0.032
Squared-Up%	$r = 0.068$ (n.s.)	+0.021

Squared-up%'s raw correlation with xwOBA is statistically indistinguishable from zero. But once bat speed, swing length, and attack angle are held constant, its standardized effect is real and substantial — about **40% of the combined bat-speed-plus-squared-up effect** on xwOBA. Bat speed remains the stronger predictor of the two, but squared-up% is far from irrelevant; it's simply invisible until the trade-off is controlled for.

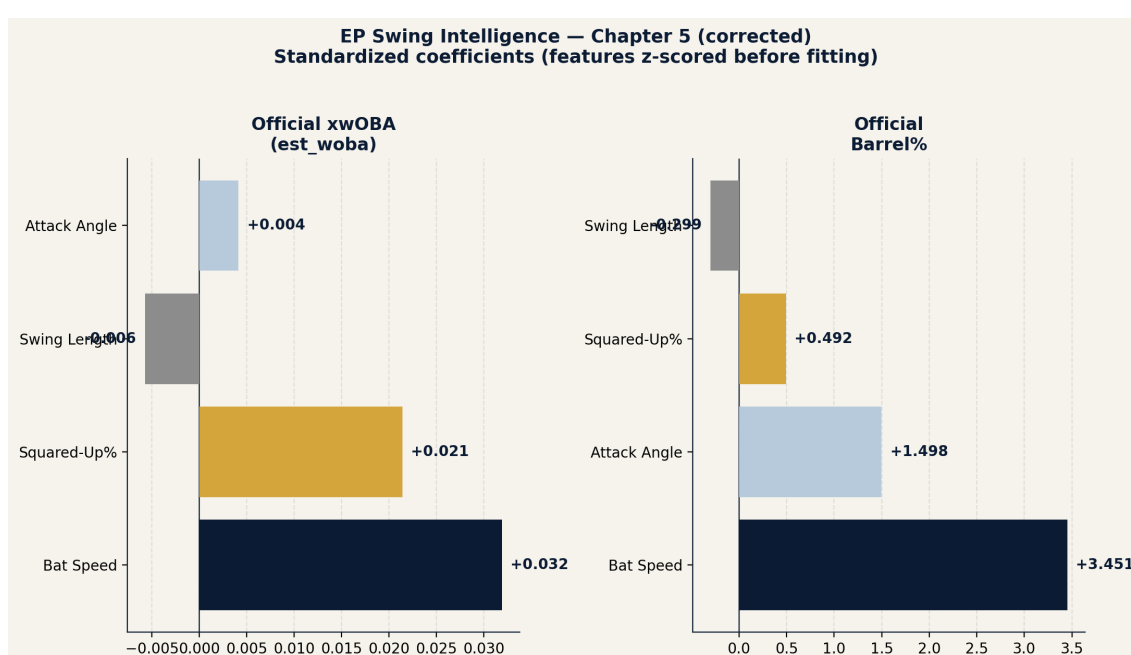


Figure 2. Standardized regression coefficients, 4-feature model ($n=205$, 10-fold CV, targets = official Savant xwOBA and Barrel%). Bat Speed leads on both targets; Squared-Up% and Attack Angle contribute meaningfully once the trade-off with Bat Speed is controlled for.

METHODOLOGICAL NOTE

An earlier pass at this analysis fit the regression on features in their native units (mph for bat speed, a 0–1 proportion for squared-up%) without standardizing first, which made squared-up%'s raw coefficient look ~40x larger purely due to its smaller numeric scale — not because it mattered more. Re-fitting on standardized (z-scored) features corrects this: model accuracy ($R^2 = 0.42$ for xwOBA, 0.63 for Barrel%) is unchanged, but the coefficients are now genuinely comparable across features.

04 — DOES THE BAT SPEED GAIN PREDICT WHAT COMES NEXT?

A Weak, Not Absent, Predictor

Isolating the 182 hitters who qualified in both 2025 and 2026, the year-over-year change in bat speed correlates with the year-over-year change in xwOBA at $r = 0.385$ ($R^2 = 0.15$) — positive, statistically significant, but leaving roughly 85% of the variance in outcome improvement unexplained by bat speed alone. This is consistent with an independent public analysis of the same question (Salorio, 2024), which found bat speed's year-to-year correlation with future contact quality ($r = 0.48$) to be the weakest among the batted-ball metrics tested — barrel rate led at $r = 0.55$.

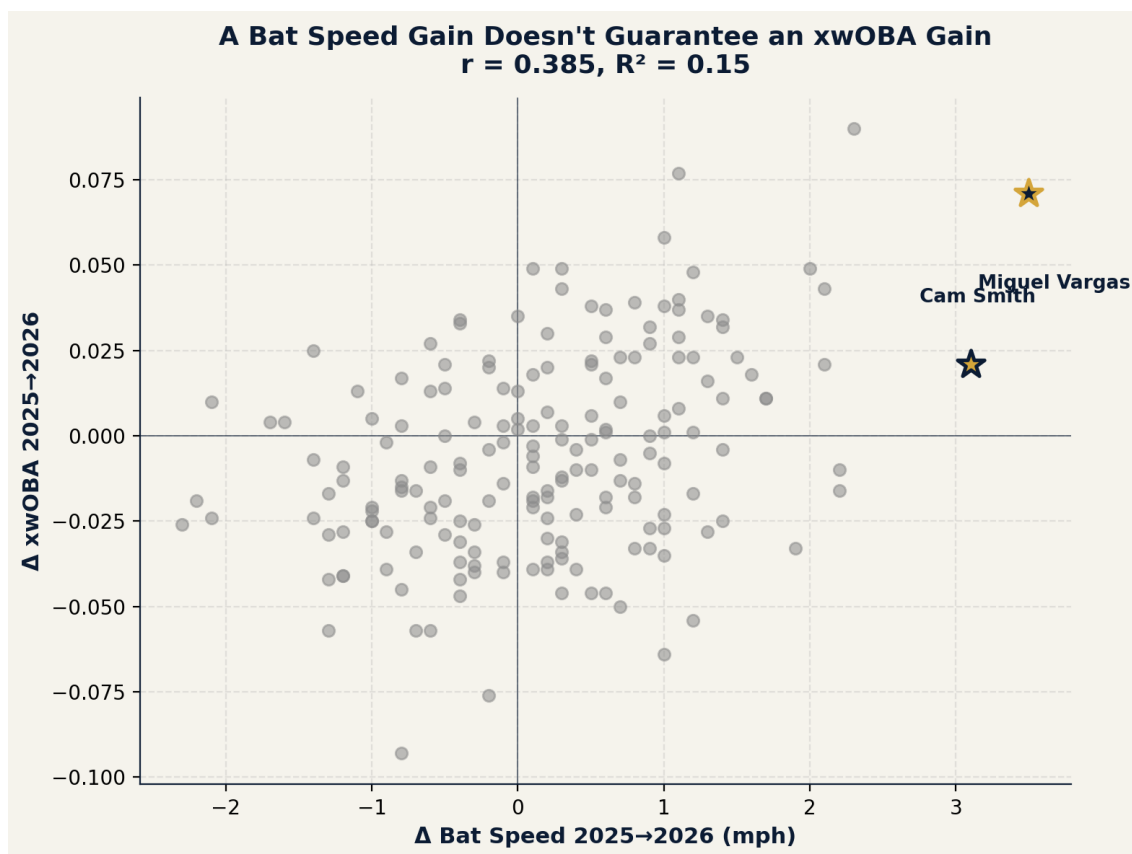


Figure 3. Δ Bat Speed vs. Δ xwOBA, 2025→2026, 182 qualified hitters. Cam Smith (#2 in bat speed gain) and Miguel Vargas (#1) started from a nearly identical bat speed jump but landed in very different places on the outcome axis.

Cam Smith ranked **#2 in all of MLB** in bat speed gain (+3.1 mph) but only **#38 of 182** in xwOBA gain (+0.021). Miguel Vargas, the **#1** bat speed gainer (+3.5 mph), posted an xwOBA gain more than three times larger (+0.071). Several of the largest bat speed gainers in the league — Fernando Tatis Jr., Mark Vientos, Daylen Lile — saw their xwOBA move *backward* despite the swing getting faster. The gain in bat speed is real and measurable; whether it converts to better outcomes depends on a second variable that bat speed alone does not capture.

05 — WHERE THE SEASON-LONG AVERAGE HIDES THE REAL PROBLEM

Pitch-Type Breakdown: The Sweeper Problem

A season average is a peak-value summary — the same limitation the VALD force plate literature warns against when it comes to relying on jump height or peak force in isolation. Decomposing Cam Smith's 2026 season by pitch type reveals exactly where the swing breaks down.

Pitch Type	wOBA	Whiff%	Run Value / 100
4-Seam Fastball	.371	23.5%	+1.1
Slider	.388	38.6%	+1.1
Sweeper	.131	51.4%	-2.3
Changeup	.206	54.8%	-0.6

Against four-seamers, Smith is a functional-to-good hitter. Against sweepers specifically, he is among the least productive hitters in the league — a .131 wOBA and 51.4% whiff rate. The season-long bat speed and squared-up% numbers are averages across all of this; they don't show that the swing's timing issue is concentrated in one specific, identifiable location: the mid-plate lateral break of a sweeper. This is the kind of detail that a single aggregate metric — no matter how favorable — will always miss.

06 — CONCLUSION

What This Case Actually Shows

Bat speed remains the stronger predictor of offensive outcome — that finding holds, corrected and confirmed, in both an external MLB sample and EP's own 205-player model. But the case against reading Cam Smith's 2026 as a pure breakout is not built on squared-up% "winning" — it's built on three independent, converging pieces of evidence: (1) his profile sits at the extreme end of a real, structural power/contact trade-off; (2) his own model-predicted xwOBA (.355) runs ahead of his actual xwOBA (.333), a negative residual even after the model already accounts for his low squared-up%; and (3) the specific pitch-type data shows exactly where the swing's timing problem lives — sweepers — rather than a diffuse, unfixable issue.

PRACTICAL READ

The bat speed gain is real and worth keeping. The 2026 season is not yet a proportional breakout, and the data points to a specific, addressable gap — recognition and timing against lateral sweep — rather than a wholesale swing problem. That is a materially different, more useful conclusion than either "he's about to break out" or "the bat speed gain doesn't matter."

Methodology. Data: Baseball Savant / Statcast Bat Tracking, Percentile Rankings, and Year-to-Year Changes leaderboards (accessed 2026). External validation sample: 177 qualified hitters, current-season bat-tracking leaderboard. EP model: 205-player 4-feature linear model (bat speed, squared-up%, swing length, attack angle → official Savant xwOBA / Barrel%, 10-fold CV), EP Swing Intelligence, Chapter 5 (corrected). Conceptual framework adapted from VALD Performance, *Practitioner's Advanced Guide to Force Plates* (discriminatory metrics, waveform-over-peak-values) and *Practitioner's Guide to Shoulders* (rate-of-development vs. peak-value distinctions). Supplementary reference: Salorio, A. (2024), "Evaluating Bat Speed's Predictive Ability."